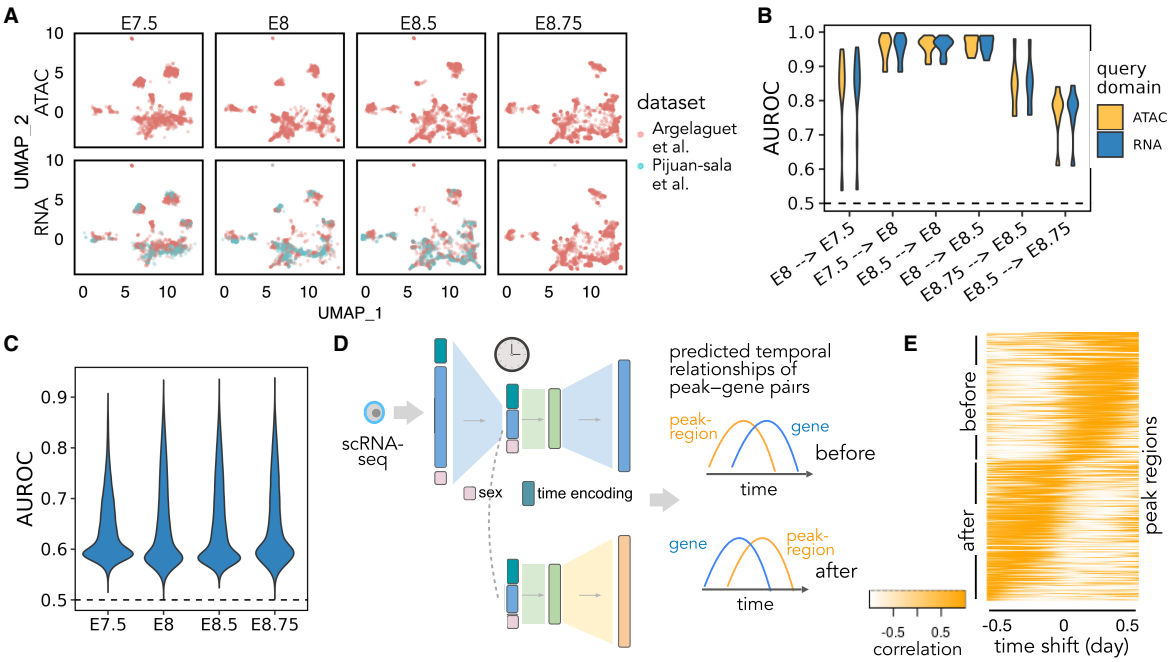

Figure 4. Multimodal temporal inference. (*A*) A UMAP embedding suggests that scRNA-seq and scATAC-seq profiles are well aligned across time and batch. Only time points with both scRNA-seq and scATAC-seq available are shown. (*B*) AUROC of the predicted differential accessibility pattern relative to those derived from the original data sets. AUROC is calculated per cell type, and differential accessibility is calculated between each held-out time point and each query time point (shown as "query time point → held-out time point"). (*C*) Peak-wise AUROC of scATAC-seq profiles predicted based on scRNA-seq relative to the original scATAC-seq profile in each held-out time point. AUROCs are calculated across all cells. (*D*) Workflow for calculating the dynamic association between peaks and genes. A query cell's scRNA-seq profile is fed into Sunbear to predict temporal patterns of gene expression and chromatin accessibility. For each pair of chromatin region and its proximal gene, we calculate the correlation coefficient between them with incremental time shifts, which results in a TLCC vector (column). (*E*) Visualization of predicted peak region–gene relationships. Heatmap of TLCC matrices on randomly selected 5000 peak regions with accessibility changes ahead of ("before") or subsequent to ("after") nearby gene expression. Peak regions are sorted based on the time shift with the maximum TLCC.

point. To do that, we used scRNA-seq profiles from the held-out time point as queries to predict each cell's corresponding chromatin accessibility profiles. We then compared the predicted profile against the original scATAC-seq measurements by comparing the predicted and true scATAC-seq profiles per peak region across all cells (Fig. 4C), and we found accurate prediction of accessibility trends in >99.9% of peaks across all time points (see the Methods subsection "Multimodal temporal inference evaluation").

Having validated Sunbear's performance on multimodal temporal inference, we trained the model on all time points, with the aim of characterizing patterns of temporal coordination between transcription and chromatin accessibility changes within each cell. Here, we focused on the hindbrain trajectory around E8.25, when the cell lineage first emerges (Qiu et al. 2022). Because there are no scATAC-seq profiles available at E8.25, we used scRNA-seq profiles collected in E8.25 in the hindbrain trajectory as queries to predict temporal cellular transcription and accessibility profiles (Fig. 4D). We then used time-lagged cross-correlation (TLCC) analysis (Chatfield 2013) to quantify the dynamic coordination between the predicted gene transcription and its nearby region's accessibility changes (Fig. 4D). Sorting genomic regions based on their temporal association pattern with nearby genes, we found diverse dynamic coordination between gene expression and chromatin priming (see the Methods subsection "Identifying temporal dynamics between chromatin priming and transcription") (Fig. 4D,E). In particular, we observe a continuous gradient of time lags between chromatin accessibilities and their nearby gene expression changes. These patterns can be categorized as peak regions changing before or after gene expression changes.

We further asked whether distinct sequence motifs are enriched in genomic regions where Sunbear predicts peak accessibilities change before and after transcription changes. This analysis showed that several transcription factor binding motifs are enriched in genome regions in the "before" category (e.g., CTCF, ZIC2, and ZIC3) (Supplemental Fig. S11) but showed none in the "after" category (for details, see the Methods subsection "Identifying temporal dynamics between chromatin priming and transcription"). Given that transcription initiation requires the opening of chromatin regions at the enhancer and promoter regions and the recruitment of transcription factors, our finding agrees with a model in which chromosomal regions tend to be primed with transcription factors before transcription changes (Bernstein et al. 2006; Rada-Iglesias et al. 2011; Lara-Astiaso et al. 2014). Among the transcription factors that we predicted to prime chromatin regions before transcriptional changes, ZIC2 and ZIC3 are important transcriptional regulators of hindbrain segmentation (Elms et al. 2003; Drummond et al. 2013). Meanwhile, CTCF is a well-known mediator for chromatin loops and has been suggested to indirectly affect the regulation of genes involved in hindbrain development (Franke et al. 2021). Overall, the above evidence suggests that Sunbear is able to identify temporal coordination between chromatin priming and transcription based on static snapshots of cells measured in one modality and that Sunbear can offer insights into transcription factors involved in lineage priming during development.